Automated Essay Scoring Using Machine Learning

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***Abstract*— Essays are frequently employed in the educational system to gauge students' comprehension of particular subjects. However, marking essays requires a lot of time and work and could be prejudiced. In order to save time, lessen human effort, and eliminate biased scoring, automated essay scoring tries to automate scoring. Due to its lack of transparency, limited language support, and requirement for tagged data for the target prompt, which is not always available, AES is still not frequently utilized. This study's goal is to examine automated essay scoring methods. The PRISMA Flow Diagram is used in this study to conduct a systematic literature review. Studies that were released between 2016 and 2021 were found. Information pertinent to the research topics is taken from these studies and then processed to provide a response. Datasets, methods, and models are found in the publications. The performance score of models utilizing the same dataset is then used to compare them. According to the study, AES uses feature engineering and deep learning as its two core methodologies. More scholars are currently researching the deep-learning methodology. CNN, LSTM, and BERT are a few examples of neural network models used in the deep learning method. Most studies use the average QWK and the ASAP dataset as performance metrics. SBLSTMA (Siamese Bidirectional LSTM Neural Network Architecture) and BERT + handcrafted-features, both with 0.801 average QWK, are the models with the highest performance score on the ASAP datasets.**

Keywords—Automated Essay Scoring, AES, Deep Learning, Feature Engineering

I. INTRODUCTION

In the Automated Essay Scoring (AES) job, valuable qualities from an essay are extracted and a numerical score is given. Essay grading takes much time, effort, and in most cases the score will be subjective towards each person. AES will reduce human workload and speed up learning feedback in education. Aside from reducing cost effectiveness, AES is considered to be more consistent and gives a more realistic score than humans.

Most AES models have been done for English essays while essays are used internationally. More research and datasets are required to develop AES in more languages. AES also still lacks transparency since most models only provide a score but not the reason the essay has that score [1]. Human scorers on the other hand can explain the score they gave and give constructive feedback to students which is important in an educational setting. AES models benefit from learning implicit (prompt-dependent) features. Implicit features are features that are identified from training on a specific labelled dataset such as semantics, on/off topic degree, and weights on topical terms [2]. However, implicit features cannot be transferred for different essay prompts. Different essays about different topics may have different vocabulary, structures, and grammatical characteristics [2]. This implies that AES model needs to train on a collection of rated essays for a specific target prompt to achieve good results, which results AES to be inapplicable when there are no rated essays available for a target prompt.

Over the past 5 years, various AES models have been proposed and AES performance scores continue to improve. Uto et.al [3] believes that further model extension might improve the accuracy further. Neural network approaches are also prone to adversarial input. Authors [4] and [5] have researched models that can handle adversarial input. However, the model performance is still not as good as using non-adversarial input.

This paper consists of 5 parts. The first part is the introduction, which contains the definition, the problems, and the challenges of the topic. The second part is theoretical background that contains the explanation of the approaches used to solve the topic. Then, the method used for the systematic review is shown in the methodology. Approaches and techniques for AES that are mostly used in the paper that we reviewed are shown in the result along with the conclusion that are reached in this literature review. All of the papers that we reviewed and considered in this literature review are cited in the references.

II. THEORETICAL BACKGROUND

Automated Essay Scoring (AES) needs to extract features from an essay to assign the appropriate score. There are two different approaches to the AES model [5] to obtain those features. The first approach is feature-engineering. This approach uses manually extracted features such as number of words, number of sentences, number or grammatical error, number of unique vocabulary, term frequency, inverse document frequency, etc. [6]. This approach benefits from explainable and flexible scoring criteria because the features are explicitly stated and can be adapted easily [5]. However, this model suffers from the lack of understanding in implicit semantic features. The models that have been used by researchers are logistic regression with SVM, e-rater technique, LSA, n-gram, and cosine similarity.

The second approach of the AES model is the neural network model (deep learning model). The model uses machine learning, specifically deep learning techniques based on word embeddings [5]. Word embedding represents the essay as vectors and using the vectors, semantic features are extracted from the process of deep learning. Neural network models generally have better performance due to their ability to extract implicit semantic features. However, it lacks understanding of hand-crafted features which is also important to determine an essay score. Some researchers [6], [5] have tried to integrate both approaches to increase AES performance. Researchers have used CNN, LSTM, and BERT as an end-to-end model.

Convolutional Neural Network (CNN) is a neural network that applies convolution layers. In AES, convolution layer can be used to extract essay representation. Dong and Zhang [7] uses CNN to extract sentence representation and uses a fully-connected hidden layer to process the sentence representation. CNN is able to cover more high-level and abstract features compared to manually handcrafted features.

Long Short Term Memory (LSTM) is a Recurrent Neural Network Model with improved capability of remembering long term memory. This model has been used for various NLP tasks resulting in great performance. LSTMs are able to extract implicit features from a sequential input and extract important semantic features in AES. In the context of AES, various researchers [8, 9, 10] words or sentences are treated as a the sequential input and the output will be the score.

Bi-directional Encoder Representations and Transformers (BERT) is a method which can generate vector representation from long sentences. Compared with traditional word embedding, BERT can essentially avoid the problem of word segmentation [2]. Unlike LSTM, BERT uses an attention mechanism to analyze sequences allowing parallelization. It can process longer input sequences and leads to faster performance. BERT is used as an end-to-end model in AES. Several authors also tried to combine BERT models with handcrafted-features which further improve performance. The authors of [6] stated that performance of the AES model employing BERT is superior than that of other feature combinations and the current ensemble-based approach [6].

III. Methodology

The PRISMA Flow Diagram is the model used in this systematic review. In our literature review, the preferred reporting items for systematic reviews and meta-analyses, or PRISMA, were employed. The PRISMA method was chosen since the method let authors ensure the complete reporting of systematic reviews and meta-analyses that had been done in the process of writing this literature review. PRISMA flow diagram in figure 1 provides an overview of the number of research papers reviewed by authors and relevant for writing this pap er.

The criteria for a research paper to be included in this review are paper that was published between 2016 and 2021, written in English, and has AES (Automated Essay Scoring) as the main topic of the paper. PRISMA diagram flow in figure 1 describes the process of the inclusion and exclusion of paper that will be used as the main baseline of the review paper. The selection process includes four steps which are:

1. Identification, which was done by searching from Google Scholar Search Engine as well as reading papers that were referenced by related papers. Searches were done by including keywords such as (1) Automated Essay Scorer, (2) Exam Assessment, (3) Essay Grading, (4) Machine Learning. Search could be done by combining keywords mentioned or adding other words with keywords. In this step, duplicate papers are removed.
2. Screening, in this step papers that are not related with the topic Automated Essay Scorer, were excluded.
3. Eligibility, papers were inspected by reading the abstract and introduction, papers that do not match or relevant with our research question were eliminated.
4. Included, full-paper reviews were conducted and research papers that do not match or relevant with our topic were eliminated leaving the included papers as our baseline.

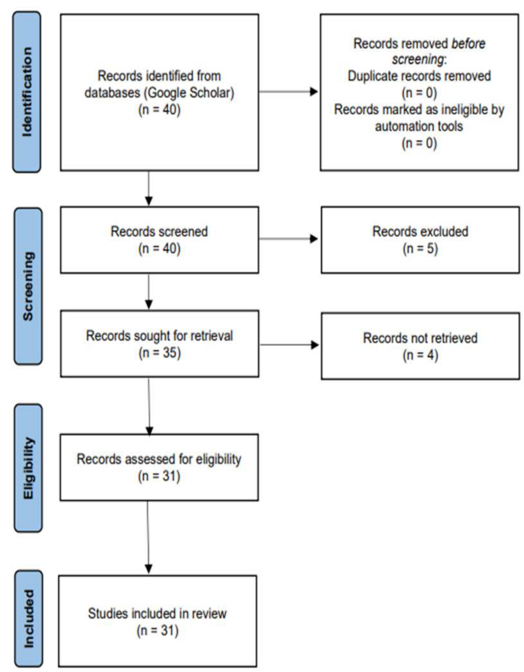

Fig. 1. PRISMA Flow Diagram

IV. RESULT

A. *What approach are most commonly used in Automated Essay Scoring?*

Based on the papers included in this review, there are two main approaches that were used for an Automated Essay Scoring. The approaches are feature engineering and deep learning. This table below contains the information about the approach used by the 31 papers included in this review.

TABLE I. PUBLICATION ABOUT THE APPROACH USED FOR AN AUTOMATED ESSAY SCORING

Approach Used	**Number of Paper**	**Paper Reference**
Deep learning model (Neural network model)	21	[1], [2], [3], [4], [6], [7], [8], [9], [10], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31]
Feature engineering model	10	[11],[12], [13], [14], [15], [16], [17], [18], [19], [20]

From the information extracted from Table 1, the most used approach for an Automated Essay Scorer is a deep learning model. Deep learning model was used by 21 out of 31 papers, which is way more than the feature engineering model. Therefore, we can conclude that deep learning is the preferred approach. Cozma et.al [11] stated that machine learning methods such as deep learning achieves better performance because it is able to capture more subtle and complex information.

Feature engineering model was used by the authors of [11, 12, 13, 14, 15, 16, 17, 18, 19, 20]. To form the document vectors, the authors of [12, 14, 18, 20] used Latent Semantic Analysis (LSA). Meanwhile, Fauzi et.al [13] used TF-IDF (Term Frequency-Inverse Document Frequency). Berggren et.al [19], Ng et.al [14], and Nguyen et.al [15] modelled AES as a regression task, and used a Support Vector Machine (SVM) to do logistic regression. Tashu et.al [17] used Word Mover's Distance to calculate the similarity between the lecturer's and the students' answers, while the authors of [13, 18, 20] used n-gram and cosine similarity. E-Rater technique was used by Ramalingam et.al [16].

B. *What deep learning techniques are used for Automated Essay Scoring?*

After reviewing relevant papers about Automated Essay Scoring, it can be concluded that deep learning is the most commonly used approach. Therefore, this paper will give a more detailed explanation on the neural networks models and algorithms that are used for Automated Essay Scoring. The models that are used are Convolution Neural Network (CNN), Long Short Term Memory (LSTM) and Bidirectional Encoder Representation from Transformers (BERT).

Convolutional Neural Network (CNN) was used by authors of [7, 24, 25]. Dong and Zhang [7] proposed the best CNN model with 0.735 QWK score. The model uses hierarchical CNN with the first convolution layer used to extract sentence representation and then a second convolution layer is used to create essay representation from the sentence vectors.

TABLE II. PUBLICATION ABOUT THE MODELS USED FOR AN AUTOMATED ESSAY SCORING

Models Used	**Number of Paper**	**Best Performance Score (average QWK on ASAP dataset)**	**Paper Reference**
LSTM LSTM LSTM+CNN LSTM-CNN-attent Bi-LSTM SBLSTMA SKIPFLOW LSTM BLA TSLF TDNN	13	0.801 (SBLSTMA [9])	[1], [2], [4], [5], [8], [9], [10], [22], [26], [28], [29], [30], [31]
BERT BERT Mobile-BERT BERT + handcrafted features	5	0.801 (BERT+handcrafted features [3])	[3], [6], [21], [23], [27]
CNN	3	0.734 (Hierarchical CNN [7])	[7], [24], [25]

Various LSTM-based models [1, 2, 4, 5, 8, 9, 10, 22, 26, 28, 29, 30, 31] were proposed for AES system. Taghipour and Ng proposed an LSTM approach without any feature engineering. The model analyses sequences of words from the essay. Dong et.al researched Attention-based Recurrent Convolutional Neural Network (LSTM+CNN+att) approach. This model treats input essays by sentence and performs attention pooling to find important words and sentences. The model performs better than LSTM. Tay et.al then proposed the SKIPFLOW LSTM model. In order to simulate the relationships between various places in an essay, the model uses parameterized tensor compositions, which produces neural coherence features that can be helpful. It can outperform LSTM by approximately 10%. predictions Liang et.al proposed a Siamese Bidirectional LSTM model. Instead of analyzing the essays only, the model can receive rating criteria information that helps the model determine distance between inputs and samples. The model outperforms SKIPFLOW and LSTM+CNN+att by approximately 5%.

By 2018, the BERT model was introduced. The BERT model tackles the weaknesses of the LSTM model and outperforms many NLP tasks. The model can achieve significant results under lower computational budget [29]. Uto et.al [3] researched feature-engineering approaches that can be applied to conventional neural networks such as BERT and LSTM. The approach is able to improve prediction of existing

AES systems without increasing complexity too much. Currently BERT has the best performance on ASAP dataset.

V. Conclusion

The PRISMA Flow Diagram is employed in this review. 40 studies regarding Automated Essay Scoring that were published between 2016 and 2021 were found using Google Scholar. Among 40 studies, five papers are excluded because they are not related with Automated Essay Scoring and four papers are not retrieved. This review included 31 studies about Automated Essay Scoring. Information pertinent to the research topics is taken from these studies and then processed to provide a response. Datasets, methods, and models are found in the publications. Performance scores are compared among models that use same dataset to identify the model with best performance.

Two main approaches that are used for AES are deep learning and feature engineering. Feature engineering relies on the use of explicit features such as number of words, sentences, grammatical errors, unique vocabulary, etc. Deep learning aims to extract implicit features using neural networks to understand semantics and topics that are relevant to the target. Over the past five years, deep learning had more research than feature engineering.

Deep learning techniques such as LSTM, CNN, and BERT are used for AES. Some studies modified deep learning approaches with feature engineering. Most of the research used ASAP (Automated Student Assessment Prize) dataset. This paper compared model bases on their score on ASAP dataset. The top performance models are SBLSTMA (Siamese Bidirectional LSTM Neural Network Architecture) and BERT + handcrafted-features with 0,801 average QWK, while the best CNN-based models are hierarchical CNN and SBLSTMA with 0.734 average QWK each.

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